Name			
USN			
Date	Conduction	Record	Viva
Allotted Marks			
Obtained Marks			

EXPERIMENT 7

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes

Aim: To create, alter, and drop databases, tables, views using Hive in Apache Ambari

Theory:

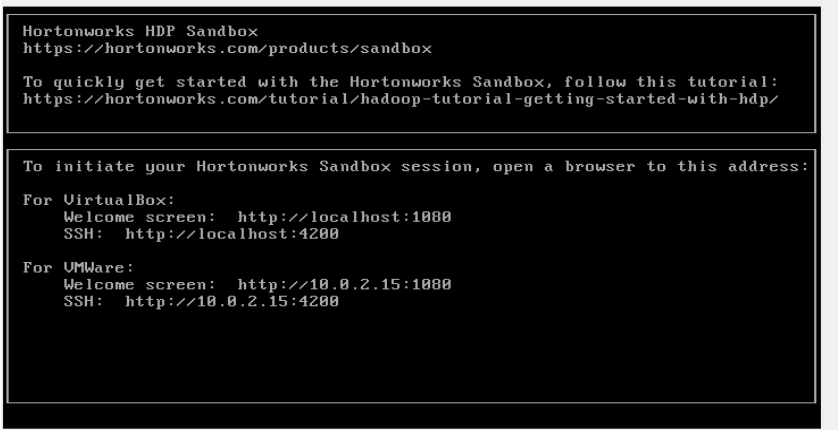
Apache Ambari is a **management and monitoring tool** for Hadoop clusters. It provides a web-based UI and RESTful APIs to deploy, configure, and manage services like **Hive, HDFS, YARN, and more**. Apache Hive is a **data warehouse** infrastructure built on top of Hadoop that allows users to query and manage large datasets using **HiveQL** (similar to SQL). It converts queries into MapReduce, Tez, or Spark jobs for execution on **HDFS (Hadoop Distributed File System).**

Using **Apache Hive on Ambari** makes it much easier to deploy, configure, monitor, and optimize Hive in a Hadoop cluster. It eliminates manual configurations and allows administrators to manage everything through an intuitive web interface.

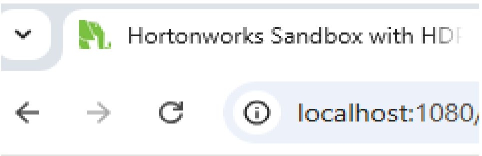
Procedure:

7.(a) To create, alter, and drop databases

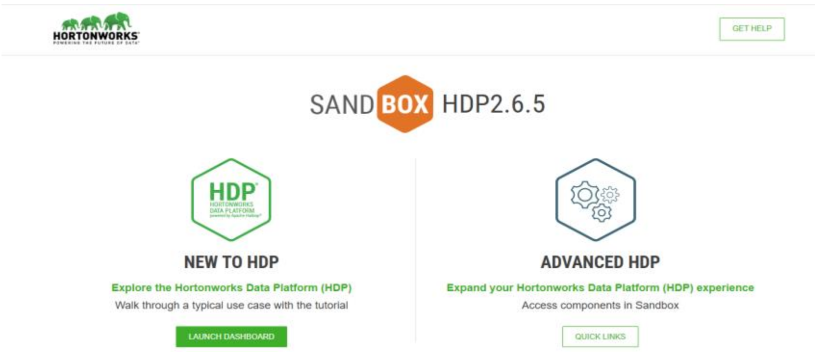
1. Start the Hortonworks virtualbox & wait for the below screen



2. go to the web browser and type localhost:1080 and press enter

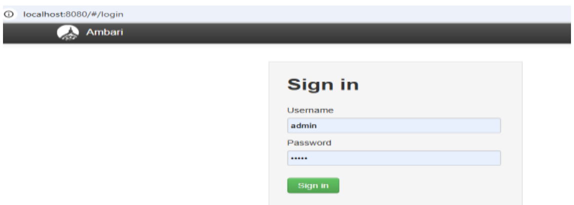


You will the below screen

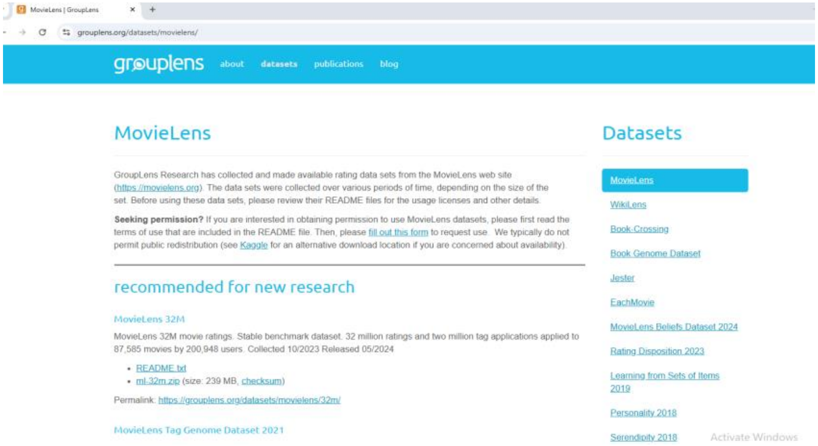


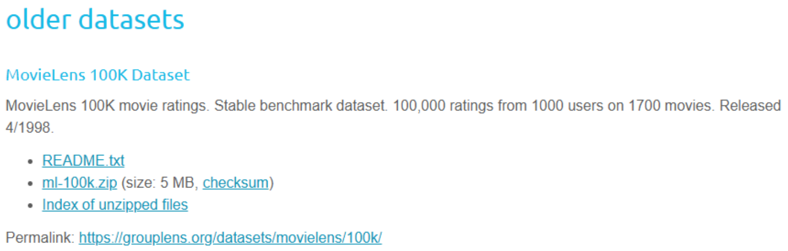
Then click Launch Dashboard

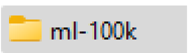
3. Next , you will be navigated to Ambari screen, Type username: admin & password: admin

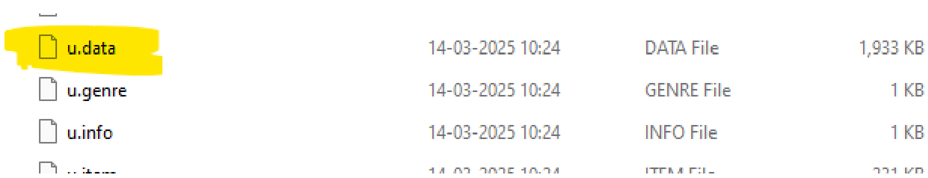


4. As we will deal with movie datasets, go to web browser and type movielens.com . Scroll through older datasets & download ml-100k.zip file & unzip the data file. Find the files u.data & u.item.







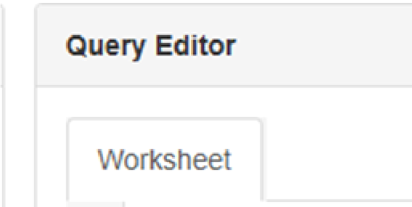


5. Now go to Ambari dashboard, grid view & select Hive view. Select the query & go to query editor .

type the following commands

here we are trying to create, alter & drop the database ; database_test

TO CREATE

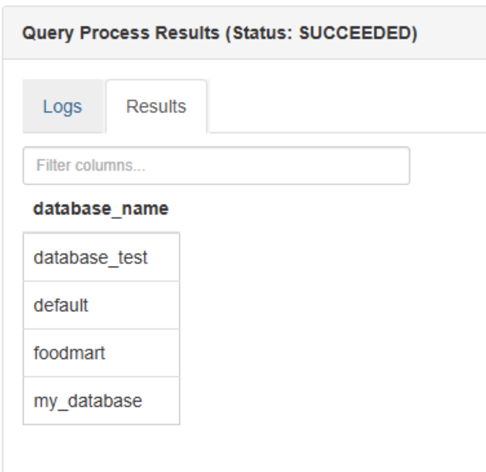


CREATE DATABASE database_test;

Execute
You will get a message as success
Now type

SHOW DATABASES;
Execute

You will see the following



You can check the details of the database by typing

Query Editor

Worksheet

```
DESCRIBE DATABASE database_test;
```

This gives the result

Query Process Results (Status: SUCCEEDED) Save results...

Logs Results

db_name	comment	location	owner_name	owner_type	parameters
database_test	""	hdfs://sandbox-hdp.hortonworks.com:8020/apps/hive/warehouse/database_test.db	admin	USER	""

TO ALTER

We are trying to alter the database by renaming the database_test to database_test1. To do so , type the following command

Worksheet *

```
ALTER DATABASE database_test RENAME TO database_test1;
```

Then you can check by typing

SHOW DATABASES;

Now check the tables existing in database_test

Worksheet

```
USE database_test;
SHOW TABLES;
```

This gives the result

Query Process Results (Status: SUCCEEDED)

Logs | Results

Filter columns...

tab_name
movie_ratings
older_movies

Now Copy a table from one database to another, using the command

Worksheet

```
USE database_test;
CREATE TABLE database_test1.older_movies LIKE database_test.older_movies;
```

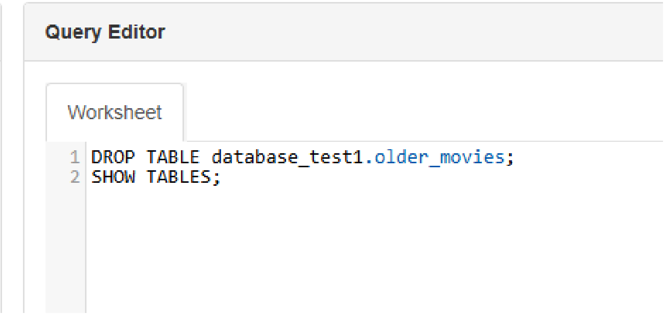
Check the table copied by typing

Worksheet

```
USE database_test1;
SHOW TABLES;
```

To drop/ delete

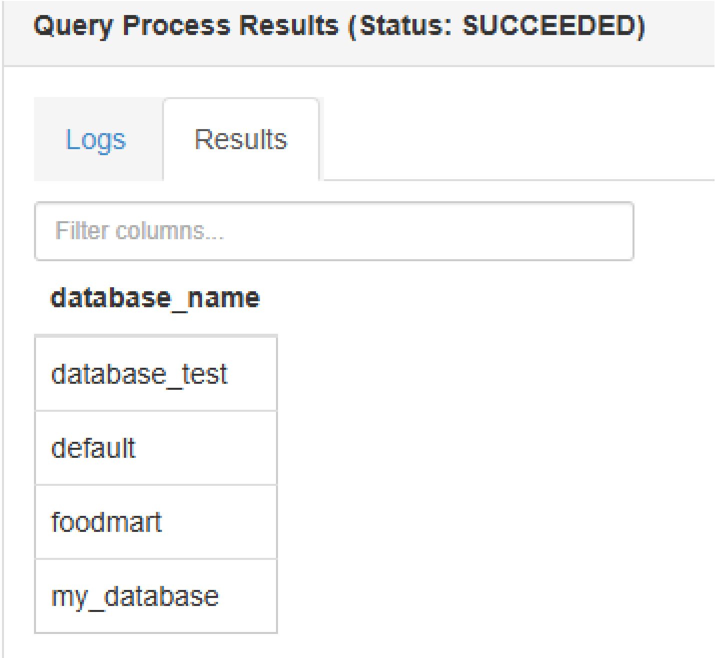
Before dropping the database, you need to delete all the tables and other objects in it



Then type

SHOW DATABASES

List of all the available databases will be shown



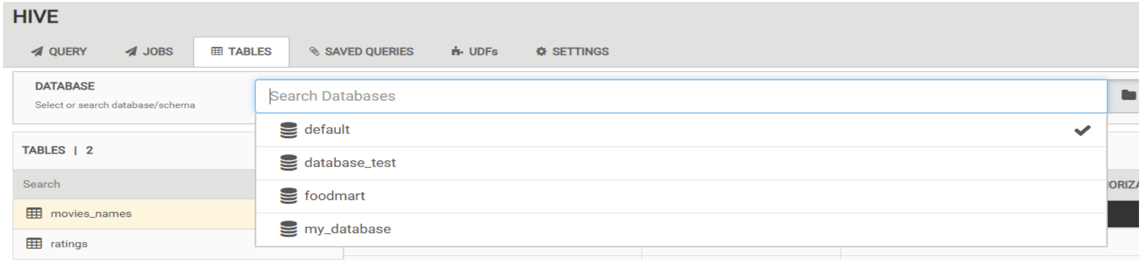
Now type

DROP DATABASE IF EXISTS database_test CASCADE;
Refresh the databases & check the list.

The `CASCADE` option will automatically delete all tables, views, and other objects within the database before dropping the database itself.

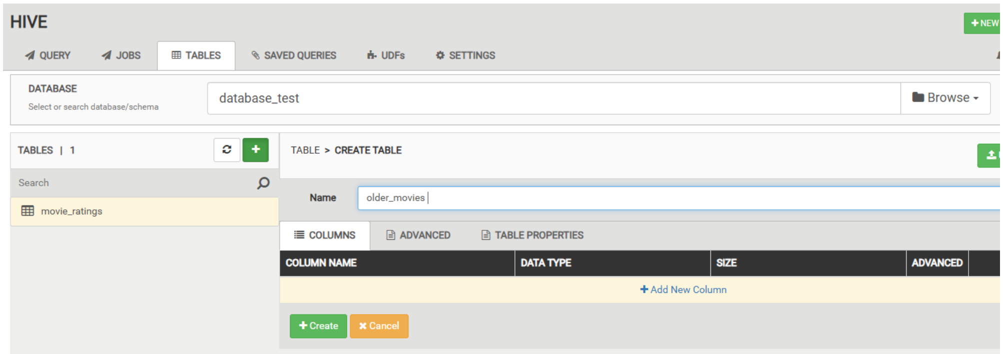
7.(b) To create, alter, and drop tables

Go to grid view & select Hive view 2.0

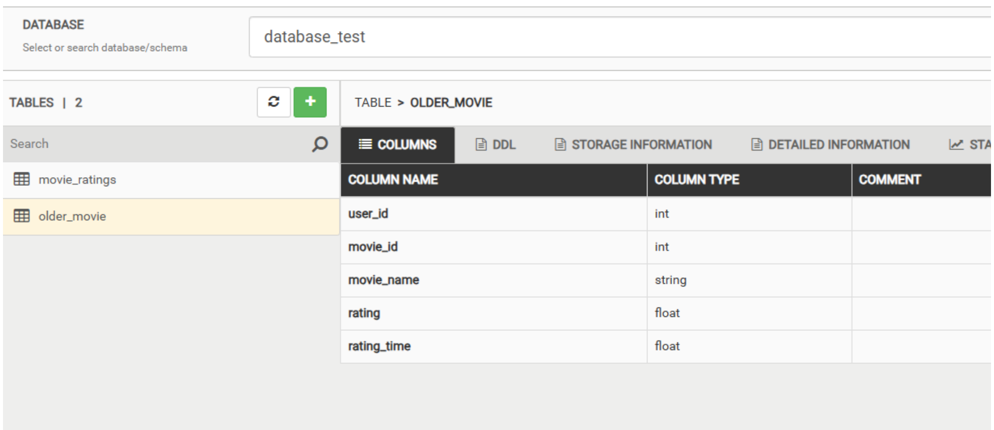


Select the database, to where you have to create a table

Click on tables & + symbol , an option of create table will appear. Give some name to the table.

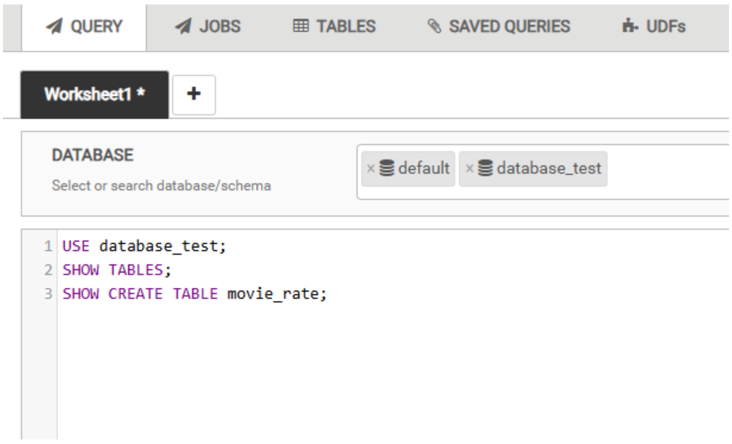


Add column names & column type, select create. A table will be created.



SQL Queries for Table Operations

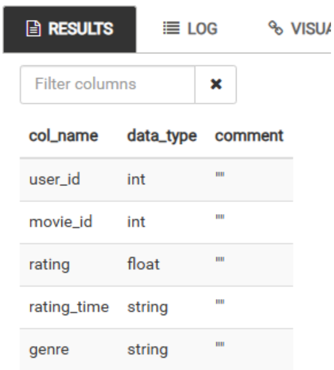
```
CREATE TABLE movie_rate (
    User_ID INT,
    Movie_ID INT,
    Rating FLOAT,
    Rating_Time STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```



To alter / add column in the table

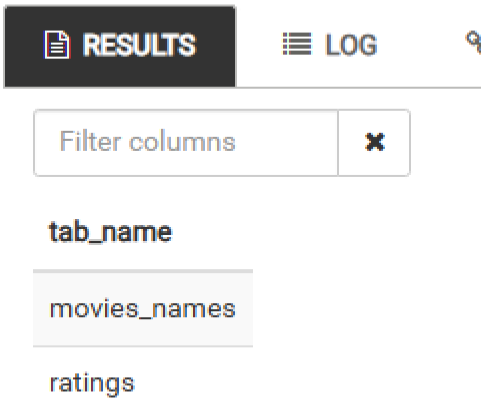
```
ALTER TABLE movie_rate ADD COLUMNS (Genre STRING);
DESCRIBE movie_rate;
```

A new entry in the table will be added



TO DROP/ DELETE tables

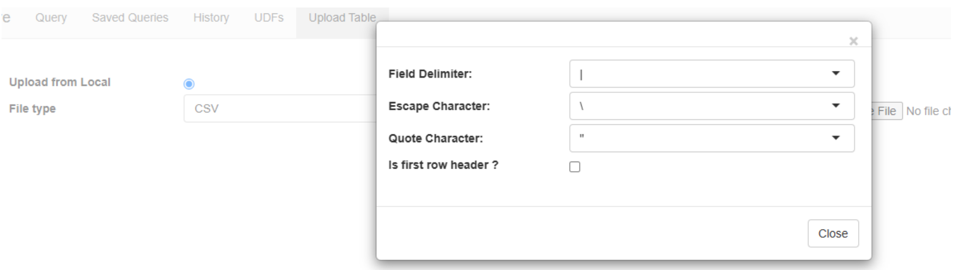
DROP TABLE movie_rate;
SHOW TABLES;



7.(c) To create, alter, and drop VIEWS

We will be using the data , which was downloaded in step 4

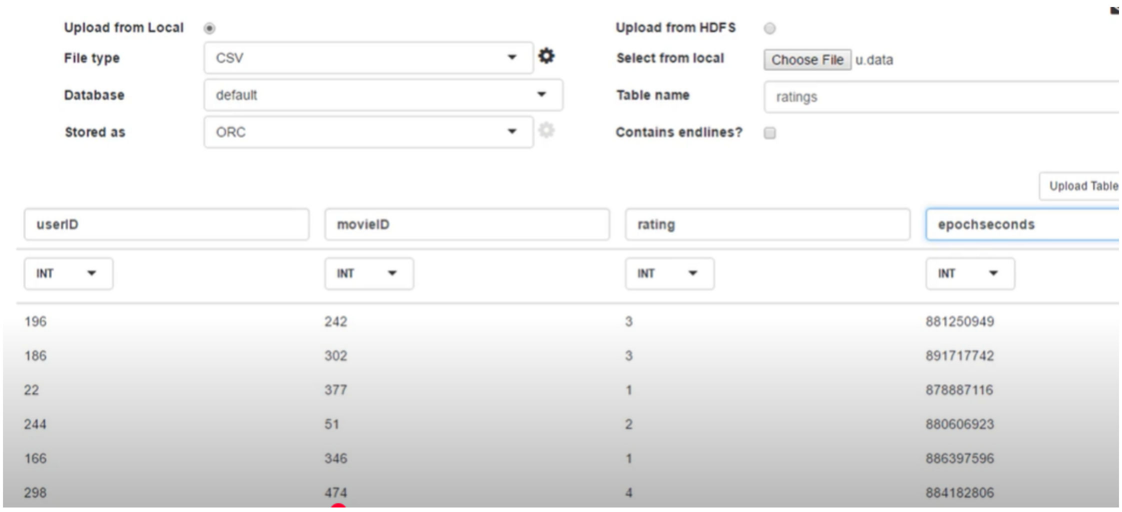
Go to grid view, Hive , upload tables



Go the settings option of file type, select Field Delimiter & select Horizontal tab.

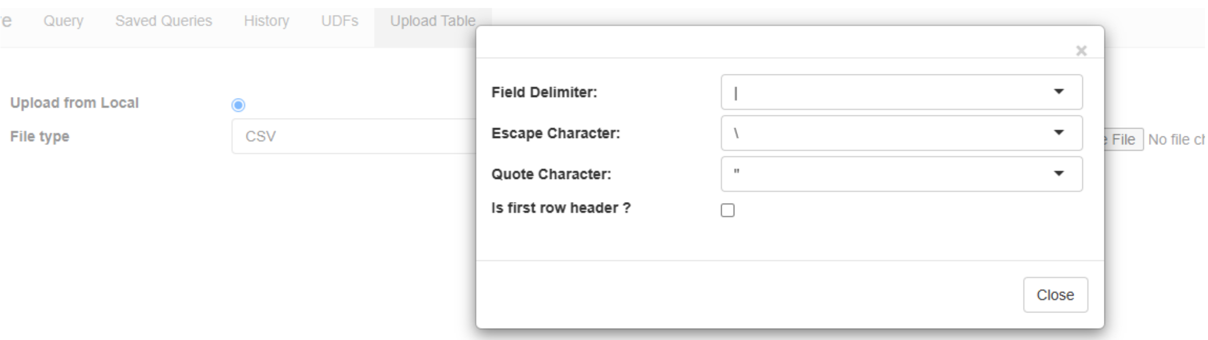
Select the data from choose file -→ select u.data. the file should appear

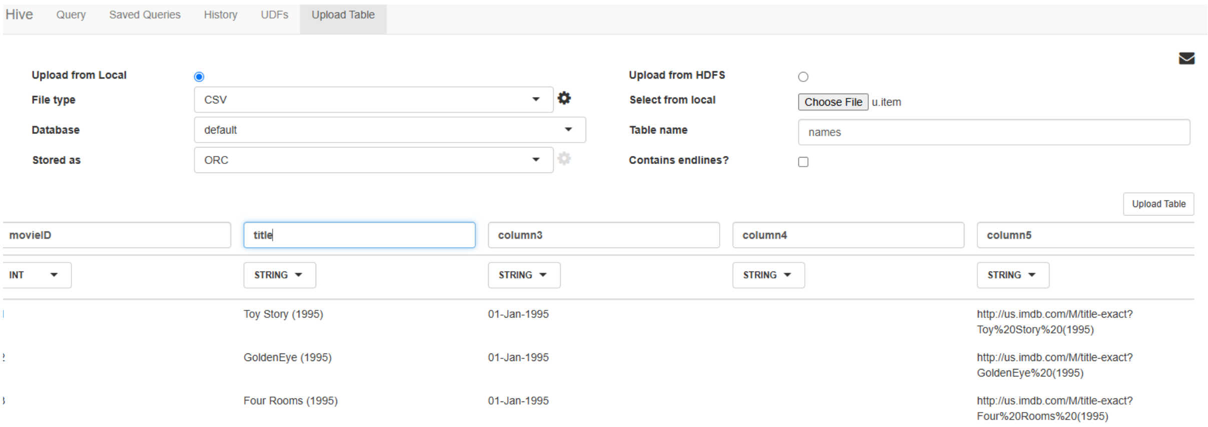
Change the column names, as given below



Repeat the same for uploading u.item

The file delimiter should be |





Once the uploading is completed, refresh the databases & check for the tables created under the default database

default
- movies_names
- names

movieid	INT
title	STRING
column3	STRING
column4	STRING
column5	STRING
column6	INT
column7	INT
column8	INT
column9	INT
column10	INT

Load more...

Once checked with the availability of tables in respective databases, type the following in the query editor

Worksheet

```
CREATE VIEW topoldMovieIDs AS
SELECT movieID, count(movieID) as ratingCount
FROM ratings
GROUP BY movieID
ORDER BY ratingCount DESC;


SELECT n.title,ratingCount
FROM topoldMovieIDs t JOIN names n ON t.movieID = n.movieID;
```

Result

Query Process Results (Status: SUCCEEDED)

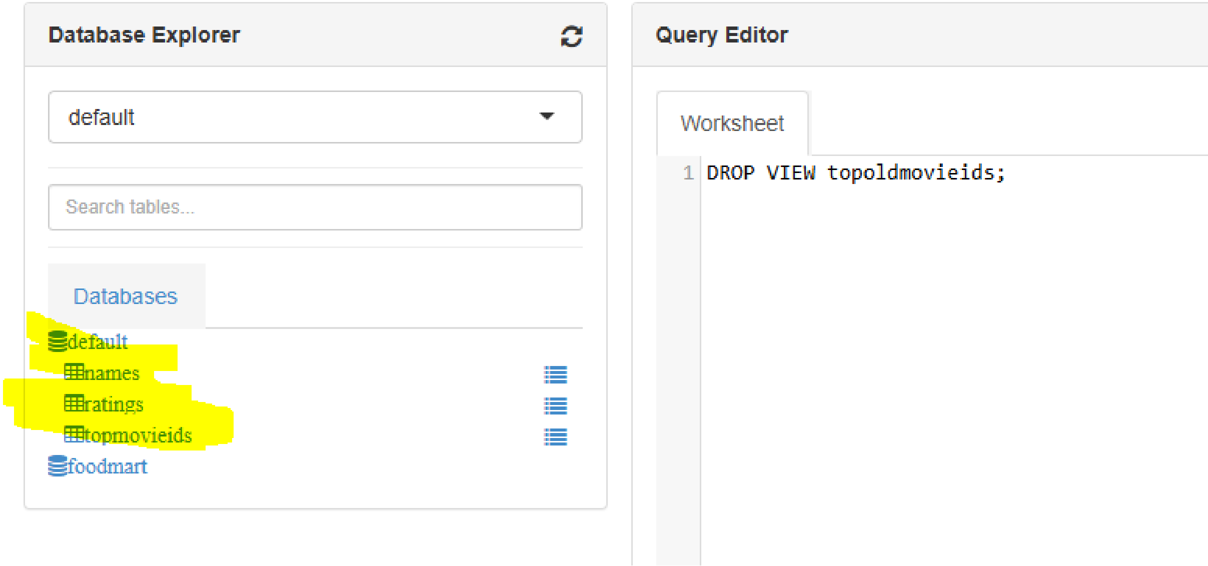
Logs Results

Filter columns...

n.title	ratingcount
Star Wars (1977)	583
Contact (1997)	509
Fargo (1996)	508
Return of the Jedi (1983)	507
Liar Liar (1997)	485
English Patient, The (1996)	481
Scream (1996)	478
Toy Story (1995)	452
Air Force One (1997)	431
Independence Day (ID4) (1996)	429
Raiders of the Lost Ark (1981)	420

Try getting the results in the reverse order.

TO DROP/DELET



Type

DROP VIEW topoldmovieids;

Check in the default database for its existence

****